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END TERM EXAMINATION
SECOND SEMESTER [B.TECH] JUNE 2025

Paper Code: BS-112

Subject: Applied Mathematics-II

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
Q1	Attempt all questions:	(5×4=20)
(a)	Prove that $f(z) = \bar{z}$ is nowhere analytic.	(5)
(b)	Find the Principal value of $\log(-1 - i)$.	(5)
(c)	Find the Laplace transformation of $(1 + t)e^{3t}$.	(5)
(d)	Taylor series expansion of $1 / (z - 2)$ in $ z < 1$ is	(5)
UNIT - I		
Q2 (a)	Find all the values of $(1 + i)^i$.	(5)
(b)	Verify that the function $u(x, y) = x(1 + y)$ is harmonic. Find its corresponding analytic function.	(5)
OR		
Q3 (a)	Evaluate the integral $\oint_C \tan z \, dz$ along the simple closed curve defined by $C : z = 5$.	(5)
(b)	Find the Line integral $\int_C (1/z) \, dz$ along the straight line with end points $z = 0$ and $z = 1 + i$.	(5)
UNIT - II		
Q4 (a)	Find the Bilinear transformation (Möbius transformation) which maps $1, -i, -1$ of the z -plane onto the points $i, -1, i$ of the w -plane respectively.	(5)
(b)	Discuss the type of singularities and find their residues for the function $f(z) = 1 / \sin z$.	(5)
OR		
Q5 (a)	Find and plot the image of the vertical line $x = 1$ as well as the horizontal line $y = 0$ under the function $f(z) = e^z$.	(5)
(b)	Prove that the integral $\int_{-\infty}^{\infty} [x^2 / ((x^2 + 1)(x^2 + 4))] \, dx = \pi / 3$.	(5)

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Q. No.	Question	Marks
UNIT - III		
Q6 (a)	Using Laplace transformation solve the ordinary differential equation: $y''(t) - 3y'(t) + 2y(t) = e^{3t}$, with $y(0) = 1$, $y'(0) = 0$.	(5)
(b)	Find the inverse Laplace transformation of $F(s) = (-11s + 25) / (s^2 - 6s + 15)$.	(5)
OR		
Q7 (a)	Determine the Fourier series of the function $f(x) = 1 - x^2$ in the interval $[-1, 1]$.	(5)
(b)	Find the Fourier transformation of $f(x) = -1$ for $ x < 1$, and 0 for $ x > 1$. Hence evaluate $\int_0^\infty [(\sin x) / x] dx$.	(5)
UNIT - IV		
Q8 (a)	Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature zero, assuming that the initial temperature is: $f(x) = x$ for $0 < x < L/2$, and $L - x$ for $L/2 < x < L$.	(5)
(b)	A tightly stretched string with fixed end points $x = 0$ and $x = l$, is initially in a position given by $y = \sin^3(\pi x / l)$. If it is released from rest from its position, find the displacement $y(x, t)$.	(5)
OR		
Q9 (a)	Solve the Laplace equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$, subject to the conditions: $u(0, y) = u(l, y) = u(x, 0) = 0$ and $u(x, a) = \sin(n\pi x / l)$.	(5)
(b)	Solve the equation $\partial u / \partial t = \partial^2 u / \partial x^2$, with boundary conditions: $u(x, 0) = 3 \sin(n\pi x)$, $u(0, t) = 0$, and $u(1, t) = 0$, where $0 < x < 1$, $t > 0$.	(5)

**** End of Question Paper ****